WORKSHEET 2.3

Estimating a Derivative from Data and Graphs

Unit 2: The Derivative

Key ideas

- From a table, estimate $f'(c)$ with the slope between the **nearest values on either side** of c. Show the difference quotient.
- From a graph, $f'(c)$ is the slope of the graph at c. At a corner the derivative does not exist.
- An estimate of a derivative carries units, and can be interpreted like any other rate.
- A known value and a known rate give a linear estimate nearby: $f(c+h) \approx f(c) + f'(c)\,h$.

Worked examples

Example 1 · From a table

The distance travelled by a train, $D(t)$ km, is recorded at times t minutes.

t	0	5	10	20
$D(t)$	0	6	15	40

Estimate $D'(7.5)$ with units.

Solution. Use $t = 5$ and $t = 10$: $\dfrac{15-6}{10-5} = 1.8$.

$D'(7.5) \approx 1.8$ **km per minute.**

Example 2 · A symmetric estimate

Given $f(2.9) = 7.1$, $f(3) = 7.5$ and $f(3.1) = 8.1$, estimate $f'(3)$.

Solution. The values on either side of 3 give $\dfrac{8.1-7.1}{3.1-2.9}$.

$f'(3) \approx 5$**.**

Example 3 · From a graph of line segments

The graph of g contains the line segment from $(1, 2)$ to $(4, 8)$. Find $g'(2)$.

Solution. On a line segment the derivative is the slope of the segment: $\dfrac{8-2}{4-1}$.

$g'(2) = 2$**.**

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Practice

Questions 1 to 7 use this table. The population of a town, $P(t)$ people, is a differentiable function of time t years.

t	0	2	4	7	10
$P(t)$	100	112	130	160	172

CORE

1. Estimate $P'(3)$.

2. Estimate $P'(8.5)$.

3. Find the average rate of change of P on $[0, 10]$.

4. State the units of $P'(t)$.

EXAM LEVEL

5. Estimate $P'(5.5)$ and interpret your answer in context.

6. Use $P(4) = 130$ and your estimate from Question 5 to approximate $P(4.5)$.

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7. Find the average rate of change of P on each of the four subintervals. What do the values suggest about how the growth of the town is changing?

8. The graph of f consists of the line segments joining $(0, 1)$, $(2, 5)$ and $(5, -1)$. Find $f'(1)$ and $f'(3)$, and explain why $f'(2)$ does not exist.

Question 9 · Free-response practice

The temperature along a metal rod, $T(x)$ degrees Celsius, is a differentiable function of the distance x centimetres from one end.

x	0	2	5	9	12
$T(x)$	80	71	62	54	51

(a) Estimate $T'(7)$. Indicate units of measure.

(b) Interpret your answer to part (a) in the context of the problem.

(c) Must there be a point of the rod at which $T'(x) = -\dfrac{29}{12}$? Justify your answer.

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Worked solutions

1. $\dfrac{130-112}{4-2} = 9$ people per year.
2. $\dfrac{172-160}{10-7} = 4$ people per year.
3. $\dfrac{172-100}{10} = 7.2$ people per year.
4. People per year.
5. $\dfrac{160-130}{7-4} = 10$. At $t = 5.5$ years the population is increasing at about 10 people per year.
6. $130 + 10(0.5) = 135$ people.
7. 6, 9, 10 and 4 people per year. The growth speeds up and then slows down, which suggests the graph of P changes from concave up to concave down.
8. $f'(1) = \dfrac{5-1}{2} = 2$ and $f'(3) = \dfrac{-1-5}{3} = -2$. At $x = 2$ the slopes from the left and right are 2 and -2: a corner.
9. (a) $\dfrac{54-62}{9-5} = -2$ degrees Celsius per centimetre.
   (b) At the point 7 cm from the end, the temperature is decreasing at about 2 degrees Celsius per centimetre of distance along the rod.
   (c) Yes. $\dfrac{T(12)-T(0)}{12} = -\dfrac{29}{12}$, and T is differentiable (so continuous) on $[0, 12]$, so the Mean Value Theorem guarantees such a point.

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